Fiscal Year:	FY 2023	**Task Last Updated:**	FY 10/24/2023
PI Name:	Pluth, Janice Ph.D.		
Project Title:	Microgravity Effects on Mammary 3D Tissue Architecture and Role of Stem Cells and Genetics		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Zip Code:	89154-9900	**Congressional District:**	1
Comments:	NOTE: Formerly at Lawrence Berkeley National Laboratory until fall 2017.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	07/26/2018	**End Date:**	07/25/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/25/2023 per NSSC information (Ed., 4/12/23) NOTE: End date changed to 7/25/2022 per NSSC information (Ed., 9/19/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Parvin, Bahram Ph.D. (University of Nevada, Reno)		
Grant/Contract No.:	80NSSC18K1464		
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Task Description:

During space travel, astronauts will experience a unique environment that includes continuous exposure to microgravity and stressful living conditions. Physiological adaptation to this is a challenge and the effect of microgravity on organ development, architecture, and function is not well understood. How microgravity may impact the growth and development of an organ is an important issue, especially as space flight becomes more commonplace. These fundamental questions can be addressed by studying simulated microgravity exposed cells in vitro. 3D tissue culture models mimicking mammary gland structures have been optimized and can be used to define if exposed cells can correctly assemble into these structures to address some of NASA's key questions regarding microgravity effects on cells and organs. Additionally, as proper activation of stem cells is key to organ formation it is important to define how microgravity may affect these specialized cells. In the proposed work we will expose a mammary epithelial cell line (Comma 1D) known to contain a high proportion of stem cells and quantify the changes in mammary stem cell populations before and after microgravity exposure. Further, we will identify how exposure to simulated microgravity affects the cells' ability to form normal 3D structures important in mammary organ development following this exposure. In addition, we will test role of genetics by isolating mouse mammary epithelial cells from mice showing increased and reduced sensitivities to cancer to define if simulated microgravity shows differential effects dependent on underlying genetics. Next, we will identify miRNA changes in stem cells isolated pre and post simulated microgravity exposure to define the underlying molecular reasons for defects in correct 3D structure formation. A final aim will be to define whether microgravity in combination with certain genetic backgrounds causes cellular changes indicative of an increased cancer risk. We hypothesize that cells exposed to simulated microgravity and assayed in 3D will allow us to define how this exposure would impact the growth and development of the mammary gland in vivo. We furthermore hypothesize that the magnitude of this response depends upon genetic susceptibility and thus will utilize cells from different mouse strains to investigate this (cancer prone and resistant). Lastly, we hypothesize that the simulated microgravity exposure in an at risk genetic background may cause changes in cells that increase the propensity to develop cancer. These studies will help identify the cellular and molecular changes and underlying genetic mechanisms of simulated microgravity and how it impacts overall tissue function.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

FINAL REPORT Award #: 80NSSC18K1464 Title of Award: Microgravity Effects on Mammary 3D Tissue Architecture and Role of Stem Cells and Genetics Period of Report: July 26, 2018 to July 25, 2023 (3-year grant with two, 1-year extensions) PI: Janice Pluth Location/address: University of Nevada, Las Vegas / 4505 S. Maryland Parkway, Las Vegas, NV 89154
Grant Objectives:

This project was designed to address how simulated microgravity (sim µg) may impact mammary gland organ formation using complex 3D tissue models. Most prior microgravity cell studies have focused on terminally differentiated cells or used immortalized cell lines, which do not provide a complete understanding of the in vivo situation. In these studies, we used a mouse cell line from a pregnant mouse, HC11, that contains higher numbers of progenitor and stem cells to better discern how stem cells may be influenced by microgravity. In addition, we used freshly isolated organoids from two very diverse strains of mice to provide a more realistic picture of microgravity effects on normal cells and to discern the role of genetics on microgravity effects. Lastly, we used 3D tissue models to have a more accurate understanding of how organ development may be impacted by microgravity.

The stated AIMS of the project:

AIM 1: To define the effects of simulated microgravity on stem cell populations in normal mouse mammary epithelial HC11 (COMMA-D1) cells in 2D and their ability to form proper 3D tissue structures.

AIM 2: To investigate the role of genetics in modulating simulated microgravity effects on mammary stem cells and their ability to differentiate and form proper 3D structures, and to exhibit changes suggesting an increased carcinogenic potential.

AIM 3: To determine the mechanistic underpinnings for simulated microgravity-dependent changes by defining changes in miRNA expression.

Overall progress/accomplishments/problems:

I think it is important to begin by saying that COVID-19 had a large impact on the ability to fully investigate the stated AIMS of this grant, even with the additional 2-year extension. In early 2020, we had several of the AIMS underway; then, in March, the lab was shut down completely. We were unable to get into the lab except for critical maintenance for that entire year and had limited access in the following year. Because people could not work, staff that had optimized the assays, and had the expertise to perform them, left for other jobs they could work at virtually. Unfortunately, issues were not resolved even after vaccines were produced and people were vaccinated. I had difficulty finding anyone to hire to replace prior staff that had even minor laboratory expertise. Thus, a new person needed to be trained on basic laboratory skills and more complicated work performed entirely by the Principal Investigator (PI). In addition, many products were unavailable or back-ordered. For example, key supplies such as growth factors and collagen were delayed for many months. Thus, the years from 2020 to 2022 allowed very minor progression on the stated aims, and back-ordered, discontinued, and delayed items remained an issue for the entire grant period. In addition to the COVID-related issues, our microgravity simulator had some breakdowns that required additional work, again stalling progress. Lastly to add to all this, we had to make a laboratory move in 2022, which obviously resulted in considerable downtime and a period where we were unable to perform experiments.

However, even amid the great challenges and difficulties, we were still able to make progress towards addressing the AIMS as noted below, generating some very interesting, publishable data. Unfortunately, if there had been more time and funds, we feel we would have been able to capitalize on the early gains made and provide even more insights on the effects of microgravity on a growing organ.

Results:

AIM 1: To define the effects of simulated microgravity on stem cell populations in normal mouse mammary epithelial HC11 (COMMA-D1) cells in 2D and their ability to form proper 3D tissue structures.

Effects of simulated microgravity on stem cell populations: Mouse mammary HC11 cells which were initiated from a pregnant mouse (and thus composed of a high number of stem cells) were used in these studies. Cells were seeded into flasks and exposed to simulated microgravity over a period of 6 days. During this exposure a portion of the normally adherent cells (ug adherent) would become non-adherent and grow in suspension (ug non Ad). We would then collect both populations to measure various characteristics.

Cell size changes with simulated µg treatment: HC11 cells were exposed to 6 days of simulated µg, then measurements of cell size were made using a cellometer. Cell sizes were then binned for each treatment population (control, µg adherent, and µg non-adherent cells). We observed a significant shift in the cell size with smaller diameter cells being increased with simulated µg treatment. Differences in cell cycle kinetics under simulated µg treatment: Cells grown under simulated µg for 6 days also revealed a difference in cell cycle profile as compared to control HC11 cells. A significantly lower percentage of G1 cells, and significantly higher level of S phase cells, were observed in the simulated µg-treated adherent group as compared to the control group.

DNA double strand breaks (DSB) increased under simulated µg treatment: HC11 cells exposed to 6 days of simulated microgravity were also collected and fixed for detection of gamma-H2AX foci as a measure of DNA DSB. Microgravity is known to produce reactive oxygen species that could potentially increase DNA DSBs; thus, levels of DNA DSBs were measured. We observed a significant increase in the levels of DNA DSBs under simulated µg treatment; but only cells that were non-adherent showed this effect, not cells treated with simulated µg that remained adherent during the treatment period.

Apoptosis levels high in simulated microgravity-treated non-adherent cells: Levels of apoptosis were increased in sim µg-exposed non-adherent populations. Cells plated in flasks were exposed for 6 days and each population then analyzed separately. Results revealed an increase in apoptosis and necrosis in the microgravity exposed, non-adherent population, and a slight increase in apoptosis in the adherent cell population as well. These effects were observed both in studies looking at sub-G1 cells using PI staining (cells with lower than 1 N content) and via staining with nexin and assay that defines both necrotic and apoptotic cells. The exposures to these cells were grown with cells in a monolayer; thus, preliminary studies were initiated to attempt to expose cells in 3D to microgravity, to compare with 2D monolayer results. Unfortunately, there was not a consistently robust way that could be optimized to expose gel-based organoid cultures to simulated microgravity and allow these to stay adherent and not damaged (if detached) during the exposure. Some preliminary findings, however, did indicate different patterns of apoptosis if microgravity exposure was to cells grown in 3D as compared to 2D. Surprisingly, in 3D, the control showed slightly higher levels of necrosis than the simulated microgravity-treated cells (5.5% vs. 2.3-1.3% for microgravity-exposed). Yet the level of apoptosis was similar between microgravity-exposed samples and control. In addition, the percent of apoptotic cells is less after growth in 3D as compared with 2D, with less than 2% of the cells in any group exhibiting apoptosis.

Growth under simulated microgravity results in an increase in a small sub-population of cells: When performing FACS analysis, it was evident that a second population appeared to be increasing when gating on FSC (forward scatter) versus SSC (side scatter). Thus, we quantified changes in this sub-population of low FSC versus low SSC population following sim µg treatment. Cells were cultured under sim µg for 6 days and then assayed for changes in FSC versus SSC cell populations. The low FSC/low FSC population made up ~12% of the total population in control and adherent cells exposed to sim µg. However, cells exposed to sim µg growing non-adherently appeared to lose much of the larger, high FSC/SSC population and have a greater percentage of the low FSC/low SSC population, representing ~50% of the total population of the cells.

Stem cell markers change with simulated microgravity exposures: Given that the HC11 mammary cell line has been reported to have a higher percentage of stem cells, we wondered if these cells were changing under sim µg treatment. Prior to treatment, most of the control cells had CD24High/CD29High markers, markers previously described by Shackleton et al. to identify mammary stem/stem cell progenitor populations. Staining cells for these markers, we observed changes following sim µg exposure. Significant changes in the percentage of CD24-/CD29- , CD24- vs. CD29+, and CD24+/CD29+ cells were observed with sim µg treatment. When small and large cell size populations based on FSC versus SSC were analyzed separately, a higher degree of change was noted in the small cell population as compared with the large cell population, with the non-adherent population showing the greatest changes in these markers as compared with the other two groups (control and sim µg-treated adherent cells).

Task Progress:

AIM 2: To investigate the role of genetics in modulating simulated microgravity effects on mammary stem cells and their ability to differentiate and form proper 3D structures, and to exhibit changes suggesting an increased carcinogenic potential.

In AIM 2, we were focused on defining if there were differences in outcome if mice of diverse genetic backgrounds were used. We compared mice that had less, and more, sensitivity to environmental exposures (C57BL/6 and BALB/c, respectively), to discern if simulated microgravity promoted changes that may indicate an increased cancer risk.

One of the 3D assays we optimized was a branching assay that mimics the ductal branching that normally occurs during mammary organ development. For these studies, organoids were isolated from the mammary gland of both strains of mice to then seed for branching studies in Corning Matrigel, upon a layer of collagen. These cultures were grown in flaskettes, filled with medium, and sealed for simulated microgravity exposure. In initial work, we were successful in initiating branching starting with organoids from the mammary gland of the C57BL/6 mice, yet unable to get organoids from BALB/c mice to branch in culture. We then performed additional studies to optimize the medium for the BALB/c mice organoids and had some success. However, COVID-related delays in obtaining growth factors and the need to shift to a different type of collagen (as our previous collagen is no longer available) made it impossible for us to replicate past successes even with the C57BL/6 organoids. At this point, we needed to isolate more organoids; so again purchased more mice to isolate additional organoids and made up all new components for the medium (as components had expired during COVID) to try this assay again. Unfortunately, even with the newer isolated organoids and medium components, we were unable to replicate earlier success and to produce branching organoids, likely because we were unable to obtain (due to shortages) all the same initial stock solutions. Thus, to date we have only been able to obtain a limited amount of 3D branching assay data, from only one strain (the C57BL/6 mouse). From this work, we observe that simulated µg appears to cause slightly shorter total lengths of the main body of the branching organoids, as compared to controls. However, interestingly, the branches were longer that emanated from these shorter organoid bodies under simulated microgravity.

Increases in a novel mammary cell population following simulated microgravity exposure, specifically in non-adherent populations: The mammary tissue is made up of both basal and luminal cell types. Specific keratins are expressed in

each cell type: K5 and K14 in basal cells, and K8 and K18 in luminal cells. In control HC11 cells they typically exhibit a K18 expression, indicative of their luminal origin. However, with exposure to simulated microgravity, the non-adherent cells show a novel increased population of cells with low/no K18 expression. This may be especially critical to further investigate, as studies have indicated that in breast cancer there is typically a loss or lowered K18 expression.

Changes in 3D Acini growth, indicative of defective growth following simulated microgravity exposure: Our studies revealed changes in 3D acini growth using mouse mammary HC11 cells following 6 days of simulated microgravity exposure. In particular, the non-adherent population of simulated microgravity-exposed cells showed a lowered density, lowered organization, and greater level of proliferation, as compared to both the control and the adherent populations. We are continuing to perform additional 3D acini experiments to ensure statistical differences can be defined, but thus far the patterns have remained consistent.

AIM 3: To determine the mechanistic underpinnings for simulated microgravity-dependent changes by defining changes in miRNA expression.

We were unable to incorporate any miRNA work during the period of this funding period, as both time and money were lacking to allow us to proceed with this work. However, this would be a very informative study, which we would like to move forward with if funds permit, as the phenotypic differences are evident as described above with microgravity exposure; and miRNA studies would likely be very informative and provide an understanding of the mechanism leading to the changes observed following simulated microgravity exposure.

We were, however, able to investigate whether there where changes in epithelial to mesenchymal transition (EMT) in cells exposed to microgravity, a process typically observed in cancer. Thus, these studies imply that this may be a mechanism for the changes that we observe under microgravity exposure.

Greater numbers of EMT-positive mammary cells following simulated microgravity treatment. Additional experiments were also performed using HC11 mouse mammary cells to answer AIM 3 goals of defining if simulated microgravity may induce changes that predict a more cancer-prone state. These studies suggest that simulate microgravity does induce changes that may be predictive of a greater cancer risk.

Epithelial-mesenchymal transition (EMT) is a biological process where epithelial cells lose their normal characteristics and instead show more mesenchymal features. For example, they will show changes in cell adhesion and polarity like cells undergoing metastasis. Changes in expression of various proteins also occur during this process, including increased expression of fibronectin and vimentin with EMT progression. To define if mammary cells exposed to simulated microgravity might have a greater potential to undergo EMT, we performed some preliminary work staining control and simulated microgravity-exposed mouse mammary cells with fibronectin to discern if there was an indication of EMT changes. This work revealed an increase in fibronectin expression with simulated microgravity exposure, as compared to control cells. Both adherent and non-adherent simulated microgravity populations showed higher levels of fibronectin as compared to controls; however, the non-adherent cells µg-treated cells revealed higher numbers of positive cells as compared to the adherent populations.

Summary and discussion of findings: In total, we have made some very significant advances and findings. The smaller diameter cells based on FSC/SSC levels we observed with sim µg exposure may point to cells becoming more stem-like; however, our experiments testing for stem cell marker changes did not appear to indicate this. Instead, we observed a decrease in the numbers of cells with classic stem cell markers in the non-adherent sim µg-treated cells, as compared to control and adherent sim µg-exposed cells. Prior to sim µg exposure, control cells showed primarily CD29+/CD24+, markers, which are typically indicative of mammary stem cells. However, following exposure to sim µg, we observed an increase in CD29-/CD24+, CD29+/CD24- and double negative cells. The increase in CD24+/CD29lo positive cells may indicate an increase in multipotent luminal progenitors that give rise to mammary structures capable of differentiating, but not able to be serially transplanted. Changes in these latter cell populations were the greatest in cells growing in solution under sim µg (non-adherent population). Thus, it is unclear what the changes in these cell markers indicate, although there is a suggestion that they may lead to changes that alter morphology and proper stem cell balance.

When sim µg-treated cells are set up and cultured in 3D matrices, cells seeded that grew in suspension under sim µg were the most affected in terms of their ability to properly grow into 3D acini structures. Cells growing in suspension under sim µg, then placed into a 3D matrix to grow into acini, show a decrease in density and organization, and an increase in the number of nuclei within the acini, as well as an increased volume. These types of changes in cells growing non-adherently under sim µg show similarities to changes that arise in tumor cells, and may indicate an increased risk for these cells to form tumors following sim µg exposure.

In summary, the findings of this work predict that developing organs exposed to microgravity in space may not properly form, and that their growth and development may be negatively impacted. In addition, the increase in proliferation and decreased organization suggests similarities to tumor phenotypes, and imply these cells may have a greater risk of developing cancer, at least for the breast mammary tissue studied. Obviously, 3D acini are not complete organs, rather a sub-structure of the mammary gland that would form in vivo; thus, further in-depth studies in animals would be useful to verify these findings and additional studies using other organ types also important to define the universality of these findings.

Bibliography (publications, meeting abstracts, etc.):

See Cumulative Bibliography.

New Technology:

There were no inventions, patents, licenses, or technologies developed as a result of this research.

References:

1 Shackleton, M. et al. Generation of a functional mammary gland from a single stem cell. Nature 439, 84-88 (2006). https://

Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Kraft A, Parvin B, Winkelmaier G, Jabbari K, Cheng Q, Pluth JM. "Effects of simulated microgravity on mammary cells in 2 and 3D tissue architecture. " ICEM 2022, International Environmental Mutagenesis and Genetics Conference, Ottawa, Ontario, Canada, Aug. 27-Sept 1, 2022. Abstracts. ICEM 2022, Ottawa, Ontario, Canada, Aug. 27-Sept 1, 2022. , Aug-2022
Abstracts for Journals and Proceedings	Macedo M, Weidmeyer B, Parvin B, Jabbari K, Chen Q, Pluth JM. "Microgravity effects on mammary stem cells and 3D tissue architecture." 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020
Abstracts for Journals and Proceedings	Kraft A, Parvin B, Winkelmaier G, Jabbari K, Cheng Q, Pluth JM. "The effects of microgravity on mammary stem cell populations in 3D tissue architecture." 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. , Feb-2022
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Articles in Peer-reviewed Journals	Winkelmaier G, Jabbari K, Chien LC, Grabham P, Parvin B, Pluth J. "Influence of simulated microgravity on mammary epithelial cells grown as 2D and 3D cultures." Int J Mol Sci. 2023 Apr 20;24(8):7615. https://doi.org/10.3390/ijms24087615 ; PMID: 37108776; PMCID: PMC10140931 , Apr-2023